Week 8: 19 Oct 2026 – 23 Oct 2026

Subject: Mathematics **Unit Title:** Algebra Foundations & Introduction to Polynomials **Grade :** 10

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents المصادر و الروابط المخصصة لأولياء الأمور
Lesson 1: Solve quadratic equations with irrational roots	• I can solve quadratic equations with irrational roots.	1.1 Flexible display of information (size, color, font, layout). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 8.1 Make goals clear, visible, and supported by short-term objectives.	Materials: Teams channel, file section>Term 1
Lesson 2: Solve quadratic equations by using quadratic formula	• I can solve quadratic equations by using quadratic formula when a=1		
Lesson 3: Solve quadratic equations by using quadratic formula	• I can solve quadratic equations by using quadratic formula when a ≠ 1.		
Lesson 4: Solve quadratic equations with complex solutions- Part 1	• I can solve quadratic equations with complex solutions.		
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	**Make sure your devices are fully charged.** **Bring your pencil case/calculator with you.**		

Technology Integration دمج التكنولوجيا	Resources shared on Teams
Cross-curricular links الروابط بين المواد الدراسية	**Numeracy**: Students operate with polynomial expressions, solve quadratic equations using multiple algebraic methods, apply the Remainder Theorem, calculate geometric series sums, and compute and interpret average rates of change — all in authentic UAE-embedded contexts. **Critical & Creative Thinking:** Students analyze real-world polynomial models, evaluate the meaning of solutions (including complex ones), and make mathematical arguments about rates of change, optimization, and cumulative growth. **Communication**: Students write mathematical reports for a non-specialist UAE audience, presenting polynomial models and their interpretations clearly and precisely. GRASPS task presentations develop oral and written mathematical communication. **Self-management:** Students manage multi-session GRASPS task work across the unit, track their own progress using project journals and exit tickets, and reflect on their learning after the summative assessment.
External Assessment Integration دمج التقييمات الخارجية	All content is covered in MAP
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	**Wellbeing:** • Confidence in solving real-world maths problems. • Resilience when learning from mistakes. • Collaboration through shared investigations and discussions. • Self-awareness by setting and reflecting on learning goals. **IB ATL Skills:** • **Thinking:** solving problems, recognising patterns, and applying proportional reasoning. • **Communication:** explaining mathematical thinking using appropriate vocabulary. • **Research:** investigating real-world examples and interpreting scale drawings. • **Self-Management:** setting goals, organising work, and managing time effectively. • **Social:** collaborating respectfully and sharing ideas with others.
Standards المعايير	N.CN.7 Solve quadratic equations with real coefficients that have complex solutions. F.IF.8a Use the process of factoring and completing the square in a quadratic function to show zeros, extreme values, and symmetry of the graph, and interpret these in terms of a context.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	How can recognizing patterns help us factor quadratic expressions efficiently? How can we transform a quadratic equation into a perfect square to find its solutions? How does the leading coefficient affect the process of completing the square? How can we represent quadratic solutions when they are not rational numbers?

Success Criteria معايير النجاح (Measurable)	Lesson 1: All students must be able to differentiate between rational and irrational solutions of quadratic equations. Most students should be able to verify if the irrational value satisfies the given quadratic equation. Some students could be able to solve word problems involving area and perimeter using quadratic equations. Lesson 2: All students must be able to apply the quadratic formula when (a=1) Most students should be able to verify their solutions by substitution. Some students could be able to solve word problems involving solve quadratic equations by applying the quadratic formula. Lesson 3: All students must be able to apply the quadratic formula when (a≠1) Most students should be able to verify their solutions by substitution. Some students could be able to solve word problems involving solve quadratic equations by applying the quadratic formula. Lesson 4: All students must be able to use the complex number i^2=-1 Most students should be able to expand and simplify expressions with i^2. Some students could be able to solve quadratic equations with complex solutions.
National Identity Domain مجالات الربط بالهوية الوطنية	Belonging الإنتماء Respect الاحترام Heritage التراث
	Students solve quadratic problems connected to **UAE development and infrastructure**, such as calculating dimensions, areas, and design measurements using factoring and completing the square Students demonstrate **respect for different mathematical approaches** by comparing factoring, completing the square, and exact radical solutions while discussing and justifying their strategies with classmates. Students apply quadratic concepts to **patterns and geometric designs inspired by UAE traditional architecture and heritage**, using factoring and quadratic equations to investigate dimensions and symmetry.